



# Reaction Wheels

LONG-LASTING AGILITY



Plug  
& play



Embedded  
electronics



Reliable  
technology

# LIGHT

series



## · Performances



**RW20**



**RW40-60**

|              |               |              |
|--------------|---------------|--------------|
| MAX MOMENTUM | 20 mNms       | 40 - 60 mNms |
| MAX TORQUE   | 2 mNm         | 6 mNm        |
| MAX SPEED    | 6,000 rpm     |              |
| ACCURACY     | Down to 1 rpm |              |

## · Mechanical characteristics

|        |                  |                 |    |                  |
|--------|------------------|-----------------|----|------------------|
| MASS   | 1 Wheel          | 0.142 kg        | 40 | 0.230 kg         |
|        | 4 Wheels Cluster | 0.850 kg        | 60 | 0.270 kg         |
| VOLUME | 1 Wheel          | Ø 48 x 28.2 mm  | 40 | Ø 67 x 44.8 mm   |
|        | 4 Wheels Cluster | 95 x 95 x 61 mm | 60 | Ø 65.6 x 46.5 mm |

## · Electrical characteristics

|                    |                                         |       |    |       |
|--------------------|-----------------------------------------|-------|----|-------|
| SUPPLY VOLTAGE     | [12 V; 16.5 V]                          |       |    |       |
| FULL PERFORMANCES  |                                         |       |    |       |
| POWER              | Steady                                  | 0.8 W | 40 | 0.9 W |
| VACUUM & MAX SPEED | Max torque                              | 3.5 W | 60 | 0.9 W |
| COM BUS            | Full duplex RS 422/485<br>-NSP PROTOCOL |       |    |       |

## · Environmental characteristics

|                        |        |                                         |         |
|------------------------|--------|-----------------------------------------|---------|
| TESTS                  | QSL    | 25 g                                    |         |
|                        | Random | 17.12 gRMS                              | 21 gRMS |
|                        | Shock  | 1,000g / 2,000Hz                        |         |
| OPERATIONAL T°         |        | - 20°C ; + 40°C                         |         |
| NON-OPERATIONAL T°     |        | - 30°C ; + 50°C                         |         |
| RADIATION              |        | < 20 k.rad                              |         |
| ELECTRONIC BOARD LEVEL |        |                                         |         |
| LIFETIME               |        | 8 years*<br>100,000 zero crossing tests |         |
| FLIGHT HERITAGE        | ✓      |                                         | ✓       |
| HIGH RELIABILITY       |        | 36 FIT                                  |         |

\*assessed under mission conditions similar to qualification

# LIGHT series · For 80 to 200 kg satellites



**RW180-500**



**RW1-2k**

## · Performances

|              |                |           |
|--------------|----------------|-----------|
| MAX MOMENTUM | 180 - 500 mNms | 1 - 2 Nms |
| MAX TORQUE   | 70 mNm         | 70 mNm    |
| MAX SPEED    | 6,000 rpm      |           |
| ACCURACY     | Down to 2 rpm  |           |

## · Mechanical characteristics

|        |                                                                   |                                        |
|--------|-------------------------------------------------------------------|----------------------------------------|
| MASS   | <b>180</b> 0.800 kg<br><b>300</b> 0.900 kg<br><b>500</b> 1.075 kg | <b>1k</b> 1.40 kg<br><b>2k</b> 1.70 kg |
| VOLUME | 100 x 100 x 46.7 mm                                               | Ø 185 x 60 mm                          |

## · Electrical characteristics

|                                                    |                                              |                                                      |
|----------------------------------------------------|----------------------------------------------|------------------------------------------------------|
| SUPPLY VOLTAGE<br><small>FULL PERFORMANCES</small> | [22 V ; 32 V]                                |                                                      |
| POWER<br><small>VACUUM &amp; MAX SPEED</small>     | <b>Steady</b> 15 W<br><b>Max torque</b> 65 W | 15 W<br>65 W                                         |
| COM BUS                                            | Full duplex RS 422/485<br>-NSP PROTOCOL      | Redundant<br>Full duplex RS 422/485<br>-NSP PROTOCOL |

## · Environmental characteristics

|                                                    |                                             |                                                 |         |
|----------------------------------------------------|---------------------------------------------|-------------------------------------------------|---------|
| TESTS                                              | <b>QSL</b><br><b>Random</b><br><b>Shock</b> | 25 g<br>21 gRMS<br>1,000g / 2,000Hz             | 21 gRMS |
| OPERATIONAL T°                                     |                                             | - 20°C ; + 40°C                                 |         |
| NON-OPERATIONAL T°                                 |                                             | - 30°C ; + 50°C                                 |         |
| RADIATION<br><small>ELECTRONIC BOARD LEVEL</small> |                                             | < 20 k.rad                                      |         |
| LIFETIME                                           |                                             | <b>10 years*</b><br>200,000 zero crossing tests |         |
| DEMISABILITY                                       |                                             | Adapted for LEO application                     |         |
| HIGH RELIABILITY                                   |                                             | 60 FIT                                          |         |

\*assessed under mission conditions similar to qualification

Traceability · process & components

**EN9100 and EIDP**

Manufacturing control

**More than 30  
parameters controlled**

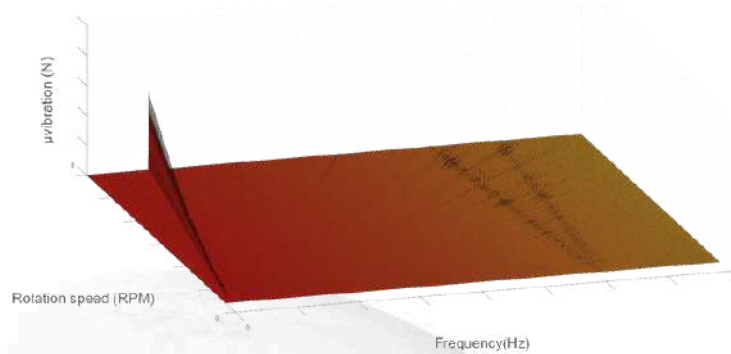


→ Not all heroes wear capes  
some **keep satellites steady.**

#### · Waterfall plot

They are developed for every reaction wheel to analyze the micro-vibrations it generates.

**Customers have the option to receive these plots at the time of delivery.**



#### · Spectrum cleaner > Earth Observation



To **enhance image stability during Earth observation** missions, Comat has developed a dedicated **damper** for the **RW40** and **RW60**.

This system significantly **reduces microvibrations induced by wheel operation**, preventing unwanted disturbances to telescope pointing accuracy and improving overall imaging performance.

#### · Photos

